A First Course in Abstract Algebra

Part 6: Extension Fields

Detailed worked solutions

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Text and numbering. John B. Fraleigh, *A First Course in Abstract Algebra*, seventh edition (2003), following the scanned textbook linked on Mathjock. A label such as 12.3 means Exercise 3 in Section 12; A.3 means Exercise 3 in the matrix appendix. Section 0 is included. These are independently written solutions; brief descriptions identify the exercises, and the textbook supplies their original statements.

How to read the solutions. Identify the assumptions and the conclusion before beginning a proof. Keep track of the underlying set, its operations, and the hypotheses of each theorem. For a calculation, follow the intermediate steps and check the result in the original definition. A counterexample must satisfy all assumptions and fail the proposed conclusion.

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29 Introduction to Extension Fields

Exercise 29.1: A shifted square root is algebraic

For $\alpha = 1 + \sqrt{2}$, we have $(\alpha - 1)^2 = 2$. Expanding gives $\alpha^2 - 2\alpha - 1 = 0$. Thus the nonzero rational polynomial $f(x) = x^2 - 2x - 1$ witnesses algebraicity over $\mathbb{Q}$.

Exercise 29.2: A sum of two square roots

For $\alpha = \sqrt{2} + \sqrt{3}$, square to obtain $\alpha^2 = 5 + 2\sqrt{6}$. Isolate the radical and square again:

$$(\alpha^2 - 5)^2 = 24.$$

Hence $\alpha^4 - 10\alpha^2 + 1 = 0$, so $f(x) = x^4 - 10x^2 + 1 \in \mathbb{Q}[x]$ proves algebraicity. For this exercise an annihilating polynomial suffices; minimality is not required.

Exercise 29.3: A shifted imaginary unit

For $\alpha = 1 + i$, the equation $(\alpha - 1)^2 = -1$ gives $\alpha^2 - 2\alpha + 2 = 0$. Therefore $x^2 - 2x + 2$ is a nonzero rational polynomial vanishing at α.

Exercise 29.4: A nested square and cube root

Let $\alpha = \sqrt{1 + \sqrt[3]{2}}$. Then $\alpha^2 - 1 = \sqrt[3]{2}$, so $(\alpha^2 - 1)^3 = 2$. Thus

$$f(x) = (x^2 - 1)^3 - 2 = x^6 - 3x^4 + 3x^2 - 3$$

is a rational annihilating polynomial. Every displayed operation is applied to the actual given number, so no converse claim about extraneous roots is needed.

Exercise 29.5: A complex nested radical

For $\alpha = \sqrt{\sqrt[3]{2} - i}$, we have $(\alpha^2 + i)^3 = 2$. Set $A(x) = (x^2 + i)^3 - 2$. Multiplying by its coefficient-conjugate gives a polynomial in $\mathbb{Q}[x]$:

$$A(x)\overline{A}(x) = x^{12} + 3x^8 - 4x^6 + 3x^4 + 12x^2 + 5.$$

The first factor vanishes at α, so the product does too. For checking the multiplication, write $A(x) = (x^6 - 3x^2 - 2) + i(3x^4 - 1)$; the product is the sum of squares of these two real polynomials.

Exercise 29.6: Minimal polynomial of a nested quadratic radical

Let $\alpha = \sqrt{3 - \sqrt{6}}$. Then $(\alpha^2 - 3)^2 = 6$, giving $p(x) = x^4 - 6x^2 + 3$. To prove minimality, first $\sqrt{6} = 3 - \alpha^2 \in \mathbb{Q}(\alpha)$. In $K = \mathbb{Q}(\sqrt{6})$, the element $3 - \sqrt{6}$ is not a square: the norm of a square $(u + v\sqrt{6})^2$ is $(u^2 - 6v^2)^2$, a rational square, but the norm of $3 - \sqrt{6}$ is $9 - 6 = 3$, not a rational square. Thus $\alpha \notin K$ and $[K(\alpha) : K] = 2$. Since $[K : \mathbb{Q}] = 2$, α has degree four. The monic annihilating quartic must therefore be $\mathrm{irr}(\alpha, \mathbb{Q}) = p(x)$.

Exercise 29.7: A rational shift inside a nested radical

For $\alpha = \sqrt{1/3 + \sqrt{7}}$, the equation $(\alpha^2 - 1/3)^2 = 7$ gives

$$p(x) = x^4 - \frac{2}{3}x^2 - \frac{62}{9}.$$

The field $\mathbb{Q}(\alpha)$ contains $K = \mathbb{Q}(\sqrt{7})$ because $\sqrt{7} = \alpha^2 - 1/3$. The norm of $1/3 + \sqrt{7}$ down to $\mathbb{Q}$ is $1/9 - 7 = -62/9$. It cannot be the norm of a square in K, since such a norm is a square of a rational number and hence nonnegative. Therefore adjoining α to K has degree two, and the full degree is four. Thus $\mathrm{irr}(\alpha, \mathbb{Q}) = p(x)$ and $\deg(\alpha, \mathbb{Q}) = 4$. Multiplying by nine gives an integer polynomial but the irreducible polynomial is conventionally monic.

Exercise 29.8: Minimal polynomial of a real-imaginary sum

Let $\alpha = \sqrt{2} + i$. Its equation $(\alpha^2 - 1)^2 = -8$ gives $p(x) = x^4 - 2x^2 + 9$. For minimality, $3/\alpha = \sqrt{2} - i$ because $(\sqrt{2} + i)(\sqrt{2} - i) = 3$. Hence

$$\sqrt{2} = \frac{\alpha + 3/\alpha}{2}, \qquad i = \frac{\alpha - 3/\alpha}{2}.$$

Thus $\mathbb{Q}(\alpha) = \mathbb{Q}(\sqrt{2}, i)$. The real field $\mathbb{Q}(\sqrt{2})$ does not contain i, so adjoining i doubles its degree two, giving degree four. Therefore p is the monic minimal polynomial and $\deg(\alpha, \mathbb{Q}) = 4$.

Exercise 29.9: The imaginary unit over the rationals

The number i satisfies $x^2 + 1 = 0$. This quadratic has no rational root (indeed no real root), so it is irreducible over $\mathbb{Q}$. Thus i is algebraic of degree two over $\mathbb{Q}$.

Exercise 29.10: A nonreal number over the reals

The number $1 + i$ satisfies $x^2 - 2x + 2 = 0$. Its discriminant is $-4 < 0$, so the polynomial is irreducible over $\mathbb{R}$. Thus $1 + i$ is algebraic of degree two over $\mathbb{R}$.

Exercise 29.11: A square root of a transcendental number

The number $\sqrt{\pi}$ is transcendental over $\mathbb{Q}$. If it were algebraic, its square would be algebraic because algebraic elements form a field (or because powers of an algebraic element lie in its finite simple extension). This would make π algebraic over $\mathbb{Q}$, contradicting the known transcendence of π.

Exercise 29.12: The same number over the reals

Since $\sqrt{\pi} \in \mathbb{R}$, the polynomial $x - \sqrt{\pi} \in \mathbb{R}[x]$ vanishes at it. Its minimal polynomial over $\mathbb{R}$ is linear and its degree is one. Algebraicity always depends on the base field.

Exercise 29.13: A square root over a rational-function field

Put $t = \sqrt{\pi}$. Since $\pi = t^2$, the base field is $\mathbb{Q}(t^2)$. The polynomial $x^2 - \pi$ shows degree at most two. It has no zero in $\mathbb{Q}(\pi)$: if a rational function $P(\pi)/Q(\pi)$ squared to π, transcendence would give the polynomial identity $P(T)^2 = TQ(T)^2$. The exponent of T on the left is even, and on the right odd, a contradiction after removing any common powers. Thus the degree is two.

Exercise 29.14: A square of pi over the rationals

The number π^2 is transcendental over $\mathbb{Q}$. If a nonzero $f \in \mathbb{Q}[x]$ satisfied $f(\pi^2) = 0$, the nonzero polynomial $f(x^2)$ would vanish at π, contradicting transcendence. Therefore no algebraic degree is assigned.

Exercise 29.15: An element already in the base field

Here $\pi^2 \in \mathbb{Q}(\pi)$, so its minimal polynomial over that field is $x - \pi^2$ and its degree is one. The coefficient $-\pi^2$ is allowed because it lies in the specified base field.

Exercise 29.16: Two different powers of a transcendental number

Set $T = \pi^3$. The element π^2 satisfies $x^3 - T^2 = 0$ over $\mathbb{Q}(T)$. A reducible cubic over a field has a root in that field. But a rational root $P(T)/Q(T)$ would imply $P(T)^3 = T^2 Q(T)^3$. Comparing exponents of the irreducible factor T gives a multiple of three on the left and an integer congruent to two modulo three on the right. This is impossible. Hence the cubic is irreducible, and π^2 has degree three over $\mathbb{Q}(\pi^3)$.

Exercise 29.17: Splitting the four-element-field quadratic

In characteristic two, $-\alpha = \alpha$ and $\alpha^2 = \alpha + 1$. Polynomial division gives

$$x^2 + x + 1 = (x + \alpha)(x + \alpha + 1),$$

since the sum of the roots is 1 and their product is $\alpha(\alpha + 1) = \alpha^2 + \alpha = 1$. Thus the second root is $\alpha + 1$.

Exercise 29.18: Constructing the nine-element field

(a) The values of $x^2 + 1$ at $0, 1, 2$ in $\mathbb{Z}_3$ are $1, 2, 2$, so it has no root and is irreducible. **(b)** With $\alpha^2 = -1 = 2$, each element is uniquely $a + b\alpha$ for $a, b \in \mathbb{Z}_3$. Use

$$(a + b\alpha) + (c + d\alpha) = (a + c) + (b + d)\alpha,$$

$$(a + b\alpha)(c + d\alpha) = (ac - bd) + (ad + bc)\alpha,$$

reducing coefficients modulo three. For example $(1+2\alpha)(2+\alpha) = 2+5\alpha+2\alpha^2 = 2\alpha$. In the requested order, the full tables are

$+$	0	1	2	α	2α	$1+\alpha$	$1+2\alpha$	$2+\alpha$	$2+2\alpha$
0	0	1	2	α	2α	$1+\alpha$	$1+2\alpha$	$2+\alpha$	$2+2\alpha$
1	1	2	0	$1+\alpha$	$1+2\alpha$	$2+\alpha$	$2+2\alpha$	α	2α
2	2	0	1	$2+\alpha$	$2+2\alpha$	α	2α	$1+\alpha$	$1+2\alpha$
α	α	$1+\alpha$	$2+\alpha$	2α	0	$1+2\alpha$	1	$2+2\alpha$	2
2α	2α	$1+2\alpha$	$2+2\alpha$	0	α	1	$1+\alpha$	2	$2+\alpha$
$1+\alpha$	$1+\alpha$	$2+\alpha$	α	$1+2\alpha$	1	$2+2\alpha$	2	2α	0
$1+2\alpha$	$1+2\alpha$	$2+2\alpha$	2α	1	$1+\alpha$	2	$2+\alpha$	0	α
$2+\alpha$	$2+\alpha$	α	$1+\alpha$	$2+2\alpha$	2	2α	0	$1+2\alpha$	1
$2+2\alpha$	$2+2\alpha$	2α	$1+2\alpha$	2	$2+\alpha$	0	α	1	$1+\alpha$

$\cdot$	0	1	2	α	2α	$1+\alpha$	$1+2\alpha$	$2+\alpha$	$2+2\alpha$
0	0	0	0	0	0	0	0	0	0
1	0	1	2	α	2α	$1+\alpha$	$1+2\alpha$	$2+\alpha$	$2+2\alpha$
2	0	2	1	2α	α	$2+2\alpha$	$2+\alpha$	$1+2\alpha$	$1+\alpha$
α	0	α	2α	2	1	$2+\alpha$	$1+\alpha$	$2+2\alpha$	$1+2\alpha$
2α	0	2α	α	1	2	$1+2\alpha$	$2+2\alpha$	$1+\alpha$	$2+\alpha$
$1+\alpha$	0	$1+\alpha$	$2+2\alpha$	$2+\alpha$	$1+2\alpha$	2α	2	1	α
$1+2\alpha$	0	$1+2\alpha$	$2+\alpha$	$1+\alpha$	$2+2\alpha$	2	α	2α	1
$2+\alpha$	0	$2+\alpha$	$1+2\alpha$	$2+2\alpha$	$1+\alpha$	1	2α	α	2
$2+2\alpha$	0	$2+2\alpha$	$1+\alpha$	$1+2\alpha$	$2+\alpha$	α	1	2	2α

Irreducibility ensures this quotient is a field, so each nonzero row of the multiplication table contains 1.

Exercise 29.19: Correcting algebraic element

An element α of an extension field E/F is algebraic over F exactly when it is a zero of a *nonzero polynomial in* $F[x]$. Both qualifications matter: the zero polynomial vanishes everywhere, and a polynomial with arbitrary coefficients in E could always be $x - \alpha$.

Exercise 29.20: Correcting transcendental element

An element $\beta \in E$ is transcendental over F exactly when no *nonzero* polynomial in $F[x]$ vanishes at β. The zero polynomial is excluded, since it vanishes at every element.

Exercise 29.21: Correcting monic polynomial

A nonzero polynomial is monic when its *leading coefficient* is 1; its other coefficients may be arbitrary. For example $x^2 - 3x + 7$ is monic although not all its coefficients are one.

Exercise 29.22: Correcting simple extension

The extension E/F is simple if $E = F(\alpha)$ for some $\alpha \in E$. Equivalently no proper subfield of E that contains *both* F *and* α exists. Omitting the condition that the subfield contain F changes the

meaning.

Exercise 29.23: True and false statements

(a) True. The transcendence of π over $\mathbb{Q}$ is a known theorem used here.

(b) True. $\mathbb{C} = \mathbb{R}(i)$.

(c) True. For $a \in F$, the nonzero polynomial $x - a \in F[x]$ vanishes at a.

(d) True. The rational field is a subfield of $\mathbb{R}$.

(e) False. A subfield has the same identity and characteristic; $\mathbb{Q}$ has characteristic zero and $\mathbb{Z}_2$ characteristic two.

(f) True. The minimal polynomial divides every rational annihilating polynomial, so their degrees are at least n.

(g) False. Take $\alpha = \sqrt{2}$, of degree two over $\mathbb{Q}$, and $f = x - \sqrt{2} \in \mathbb{R}[x]$, of degree one.

(h) True. Kronecker's construction adjoins a zero using a quotient by an irreducible factor.

(i) False. $x^2 + 1 \in \mathbb{R}[x]$ has no zero in the extension $\mathbb{R}$ itself.

(j) True. Evaluation $\mathbb{Q}[x] \to \mathbb{Q}[\pi]$ is onto and is injective by transcendence of π.

Exercise 29.24: Making transcendental numbers algebraic over larger fields

(a) Take $F = \mathbb{Q}(\pi^3)$. The polynomial $x^3 - \pi^3$ vanishes at π, and has no root in F: treating $T = \pi^3$ as an indeterminate, $P(T)^3 = TQ(T)^3$ contradicts exponents of T. A cubic with no root is irreducible. Hence the degree is three. **(b)** Take $E = \mathbb{Q}(e^{10})$. The element $u = e^2$ is transcendental over $\mathbb{Q}$, since otherwise e would be algebraic over $\mathbb{Q}$. For any transcendental u, put $L = \mathbb{Q}(u^m)$ and let A be the L-span of $1, u, \ldots, u^{m-1}$. These powers are independent: after clearing denominators in a proposed relation, powers occurring in different terms have different exponent residues modulo m, so no nonzero polynomial relation can vanish at the transcendental u. Thus $\dim_L A = m$. Reducing powers using $u^m \in L$ shows that A is closed under multiplication and contains 1. For $0 \neq a \in A$, multiplication by a is an injective linear map $A \to A$, because A lies in the field $\mathbb{Q}(u)$. Finite dimension makes it surjective, so $ab = 1$ for some $b \in A$. Hence A is a field. It contains L and u and is contained in $\mathbb{Q}(u)$, so $A = \mathbb{Q}(u)$ and $[\mathbb{Q}(u) : \mathbb{Q}(u^m)] = m$. Taking $m = 5$ proves e^2 has degree five over E.

Exercise 29.25: Splitting an irreducible binary cubic

(a) The cubic $p = x^3 + x^2 + 1$ takes value 1 at both elements $0, 1$ of $\mathbb{Z}_2$, so has no linear factor and is irreducible. **(b)** If $p(\alpha) = 0$, then $\alpha^3 = \alpha^2 + 1$. Squaring an equation in characteristic two shows that α^2 and α^4 are roots too. They are distinct: α has multiplicative order seven in the field of eight elements, so its first, second, and fourth powers differ. Reducing powers gives $\alpha^4 = \alpha^2 + \alpha + 1$. Thus

$$p(x) = (x + \alpha)(x + \alpha^2)(x + \alpha^2 + \alpha + 1).$$

The first long division gives $p(x) = (x + \alpha)(x^2 + (1 + \alpha)x + \alpha^2 + \alpha)$; the other two displayed roots factor that quadratic. This supplies the actual linear factors, not just their existence.

Exercise 29.26: The additive and multiplicative groups of eight elements

Degree three over $\mathbb{Z}_2$ gives the unique representation $a + b\alpha + c\alpha^2$ with binary coefficients. Addition is coordinatewise, so the additive group is $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$. There are seven nonzero elements. Their multiplicative group has prime order seven, so any nonidentity element generates it by Lagrange's theorem. Hence it is $\mathbb{Z}_7$. The additive group is not cyclic: every nonzero element has additive order two.

Exercise 29.27: Why minimal polynomial is an appropriate name

Division by the monic irreducible polynomial $p = \operatorname{irr}(\alpha, F)$ shows that any $f \in F[x]$ with $f(\alpha) = 0$ is a multiple of p. Indeed its remainder would have smaller degree and still vanish at α, so must be zero. Thus p has the smallest possible degree among nonzero annihilating polynomials, and monicity makes it unique.

Exercise 29.28: Synopsis of Kronecker construction

Choose an irreducible factor p of the given nonconstant $f \in F[x]$ and form the field $E = F[x]/(p)$. The constant cosets embed F in E, and the coset $\alpha = x + (p)$ satisfies $p(\alpha) = 0$, hence $f(\alpha) = 0$ because p divides f.

Exercise 29.29: Exchanging a transcendental generator

Since α is algebraic over $F(\beta)$, clear denominators in an annihilating equation to obtain

$$H(\alpha, \beta) = 0, \qquad H(X, Y) \in F[X, Y] \setminus \{0\}.$$

Here clearing denominators can be done without making every coefficient zero when evaluated at β; start with actual nonzero coefficients in $F(\beta)$ and choose polynomial representatives. Regard $H(\alpha, Y)$ as a polynomial over $F(\alpha)$. It is not the zero polynomial: some coefficient is a nonzero polynomial in α, and transcendence of α prevents its vanishing. Its value at $Y = \beta$ is zero. Its degree in Y cannot be zero either, because that would give a nonzero element equal to zero. Thus β is algebraic over $F(\alpha)$.

Exercise 29.30: Counting a finite simple extension

If $\operatorname{irr}(\alpha, F)$ has degree n, every element of $F(\alpha)$ has a unique expression

$$a_0 + a_1\alpha + \cdots + a_{n-1}\alpha^{n-1}, \qquad a_i \in F.$$

Spanning follows from division by the minimal polynomial, and uniqueness from its minimal degree. Each of the n coefficients has q choices, independently. Thus there are exactly q^n elements.

Exercise 29.31: A twenty-seven-element field

(a) The polynomial $x^3 + 2x + 1$ takes value 1 at each of $0, 1, 2$ in $\mathbb{Z}_3$, so has no root and is irreducible.
(b) Therefore $\mathbb{Z}_3[x]/(x^3 + 2x + 1)$ is a field. Each coset has a unique representative $a + bx + cx^2$ with $a, b, c \in \mathbb{Z}_3$, giving $3^3 = 27$ elements.

Exercise 29.32: Quadratic extensions in every prime characteristic

(a) For odd p, the nonzero residues pair as $\{a, -a\}$, with two distinct members per pair. Their squares agree, and equality of squares implies this same pairing because $(a-b)(a+b) = 0$ in the field. Thus there are only $(p-1)/2$ nonzero squares, fewer than $p-1$, so a nonsquare exists. **(b)** Choose a nonsquare $d \in \mathbb{Z}_p$. Then $x^2 - d$ has no root and is irreducible, and its quotient field has p^2 elements. For the omitted prime $p = 2$, use $x^2 + x + 1$, which has no root in $\mathbb{Z}_2$ and produces four elements. This covers every prime.

Exercise 29.33: Nonconstant rational functions stay transcendental

Let $\beta = f(\alpha)/g(\alpha) \notin F$, where $f, g \in F[x]$ and $g(\alpha) \neq 0$. The polynomial $f(x) - \beta g(x) \in F(\beta)[x]$ vanishes at α. It is nonzero, since otherwise any nonzero coefficient of g would show $\beta \in F$. Thus α is algebraic over $F(\beta)$. If β were algebraic over F, adjoining it gives a finite extension, and then adjoining α gives another finite extension. Their composite would make α algebraic over F, contradicting its transcendence. Hence β is transcendental over F.

Exercise 29.34: A cubic-radical field without checking every axiom

The polynomial $x^3 - 2$ is irreducible over $\mathbb{Q}$ by Eisenstein's criterion at 2. Therefore $\mathbb{Q}(\sqrt[3]{2})$ has basis $1, \sqrt[3]{2}, (\sqrt[3]{2})^2$ and consists exactly of the displayed expressions with rational coefficients. Since it is a simple algebraic extension, it is already a field. This identifies the set as a subfield of $\mathbb{R}$ without separately proving formulas for inverses.

Exercise 29.35: Fields of orders eight, sixteen, and twenty-five

For order eight use $\mathbb{Z}_2[x]/(x^3 + x + 1)$; the cubic has no binary root, so is irreducible. For order sixteen use $\mathbb{Z}_2[x]/(x^4 + x + 1)$. It has no binary root. A reducible quartic with no linear factor would be a product of irreducible quadratics, but the only monic irreducible binary quadratic is $x^2 + x + 1$. Modulo that quadratic, $x^3 = 1$ and $x^4 = x$, so $x^4 + x + 1$ has remainder 1 and is not divisible by it. Hence it is irreducible. For order twenty-five use $\mathbb{Z}_5[x]/(x^2 + 2)$; a root would have square 3, which is not among $0, 1, 4$. Unique remainders of degrees less than $3, 4, 2$ give respectively $2^3, 2^4, 5^2$ elements.

Exercise 29.36: Every element of a finite field is algebraic

Let $|F| = q$. Its nonzero elements form a group of order $q-1$, so Lagrange's theorem gives $a^{q-1} = 1$ for every $a \neq 0$. Thus all elements, including zero, satisfy $a^q - a = 0$. The nonzero polynomial $x^q - x$ has coefficients in the prime subfield $\mathbb{Z}_p$, so every element is algebraic over that subfield.

Exercise 29.37: Prime-power order

List the finitely many elements as $a_1, \ldots, a_r$ and form the chain $F_0 = \mathbb{Z}_p$, $F_i = F_{i-1}(a_i)$. By Exercise 36, each a_i is algebraic even over $\mathbb{Z}_p$, hence also over F_{i-1}. If its degree there is d_i, Exercise 30 gives $|F_i| = |F_{i-1}|^{d_i}$. Starting from $|F_0| = p$ and iterating yields $|F| = p^{d_1 \cdots d_r}$, a prime power. Elements already adjoined simply contribute $d_i = 1$.

30 Vector Spaces

Exercise 30.1: Three disjoint bases

Use the three sets $B_1 = \{(1,0),(0,1)\}$, $B_2 = \{(2,0),(0,2)\}$, and $B_3 = \{(3,0),(0,3)\}$. No vector appears in two sets. Each spans $\mathbb{R}^2$ because $(x,y) = (x/k)(k,0) + (y/k)(0,k)$ for $k = 1,2,3$. The same equation has unique coefficients, so each pair is independent and is a basis.

Exercise 30.2: Testing a symmetric triple

Put the vectors into the columns of

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}.$$

Its determinant is $1(0-1) - 1(1-0) = -2 \neq 0$, so its columns are independent and form a basis of $\mathbb{R}^3$. More explicitly, solving $a+b=x$, $a+c=y$, $b+c=z$ gives uniquely $a = (x+y-z)/2$, $b = (x+z-y)/2$, $c = (y+z-x)/2$. This also directly verifies spanning and uniqueness.

Exercise 30.3: A dependent triple

Name the first two vectors $u = (-1,1,2)$ and $v = (2,-3,1)$. Then

$$-2u + 4v = (2,-2,-4) + (8,-12,4) = (10,-14,0),$$

the third vector. Therefore the triple is linearly dependent and is not a basis of $\mathbb{R}^3$. This conclusion does not require checking spanning separately.

Exercise 30.4: A quadratic rational extension as a vector space

A basis for $\mathbb{Q}(\sqrt{2})$ over $\mathbb{Q}$ is $\{1, \sqrt{2}\}$. The polynomial $x^2 - 2$ is irreducible by irrationality of $\sqrt{2}$, so reduction by it gives spanning by these two powers. If $a + b\sqrt{2} = 0$ with rational a, b and $b \neq 0$, it would make $\sqrt{2}$ rational; hence $a = b = 0$.

Exercise 30.5: A radical already in the base field

Since $\sqrt{2} \in \mathbb{R}$, adjoining it changes nothing: $\mathbb{R}(\sqrt{2}) = \mathbb{R}$. A basis over $\mathbb{R}$ is $\{1\}$, because every real number is a unique real multiple of 1.

Exercise 30.6: A cubic rational extension as a vector space

A basis is $\{1, \sqrt[3]{2}, (\sqrt[3]{2})^2\}$. Indeed $x^3 - 2$ is irreducible over $\mathbb{Q}$ by Eisenstein at 2, so the simple extension has degree three and the first three powers form a basis. Every higher power reduces using $(\sqrt[3]{2})^3 = 2$.

Exercise 30.7: Complex numbers over the reals

The set $\{1, i\}$ is a basis for $\mathbb{C}$ over $\mathbb{R}$. Every complex number is $a + bi$ with $a, b \in \mathbb{R}$, giving spanning, and equality $a + bi = 0$ forces both its real and imaginary parts to vanish, giving independence.

Exercise 30.8: Gaussian rationals as a vector space

The set $\{1, i\}$ is also a basis for $\mathbb{Q}(i)$ over $\mathbb{Q}$. The polynomial $x^2 + 1$ is irreducible over $\mathbb{Q}$, so all elements are uniquely $a + bi$ for rational a, b. The ambient field here is smaller than $\mathbb{C}$, but the basis argument is the same.

Exercise 30.9: A fourth-root extension

Let $t = \sqrt[4]{2}$. By Eisenstein at 2, $x^4 - 2$ is irreducible over $\mathbb{Q}$. Hence $\{1, t, t^2, t^3\}$ is a basis for $\mathbb{Q}(t)$ over $\mathbb{Q}$. The relation $t^4 = 2$ reduces higher powers, and a dependence of these four powers would give an annihilating polynomial of degree less than four, contradicting irreducibility.

Exercise 30.10: The other nontrivial element of the four-element field

Let $\beta = 1 + \alpha$, where $\alpha^2 + \alpha + 1 = 0$ in characteristic two. Then $\beta^2 = 1 + \alpha^2 = \alpha$, and

$$\beta^2 + \beta + 1 = \alpha + (1 + \alpha) + 1 = 0.$$

Neither 0 nor 1 is a root of $x^2 + x + 1$ in $\mathbb{Z}_2$, so the polynomial is irreducible. Thus $\operatorname{irr}(1 + \alpha, \mathbb{Z}_2) = x^2 + x + 1$.

Exercise 30.11: Correcting spanning

A subset S spans V if every vector of V can be expressed as a *finite* linear combination of elements of S. Uniqueness is not required. For example $\{(1, 0), (0, 1), (1, 1)\}$ spans $\mathbb{R}^2$ but permits multiple expressions for $(1, 1)$.

Exercise 30.12: Correcting linear independence

The vectors in S are linearly independent if the *only* finite linear combination of distinct vectors of S equal to zero has every coefficient zero. Zero always has the trivial expression with all coefficients zero; what independence excludes is a nontrivial one.

Exercise 30.13: Dimension as a minimum spanning size

The proposed definition is correct for a finite-dimensional vector space. A basis spans, so its size is an attainable spanning size. Conversely deleting redundant elements from any finite spanning set yields a basis, so no spanning set can have fewer elements than a basis. This minimum equals the usual dimension.

Exercise 30.14: Correcting basis

A basis is a subset that spans the vector space and is linearly *independent*. Replace "dependent" by "independent." The two requirements ensure respectively existence and uniqueness of coordinates.

Exercise 30.15: True and false statements

(a) True. Addition is a binary operation on V.

(b) False as a general assertion. Scalars add in F, and F need not be the vector set V; for example in $\mathbb{R}^2$ a real scalar is not an ordered pair. Particular realizations may identify some scalars with vectors.

(c) True. A field is closed under multiplication.

(d) True. Scalar multiplication is a map $F \times V \to V$.

(e) False. $F[x]$ has no finite spanning set: the degrees in finitely many polynomials are bounded.

(f) False. A basis is independent by definition.

(g) False. A set containing zero has the nontrivial relation $1 \cdot 0 = 0$.

(h) True. α^2 lies in the finite extension $F(\alpha)$, whose elements are algebraic over F.

(i) True. The same theorem applies to $\alpha + \alpha^2 \in F(\alpha)$.

(j) True under the usual axiom of choice. Zorn's lemma gives a maximal independent subset, which must span or another vector could be adjoined.

Exercise 30.16: Subspaces and their intersection

(a) A subspace is a subset $W \subset V$ that is itself a vector space over the same field under the inherited operations. Equivalently $0 \in W$, and $u + v$ and cu belong to W whenever $u, v \in W, c \in F$. **(b)** Every subspace contains zero, so their intersection does. A sum of two vectors in every subspace lies in every subspace, as does any scalar multiple. Hence the intersection is a subspace. An empty intersection is V, which also works.

Exercise 30.17: The span generated by a subset

(a) Define $\mathrm{Span}(S)$ as the intersection of all subspaces containing S. This family is nonempty because V itself belongs to it. The intersection is a subspace by Exercise 16 and is the smallest such subspace. **(b)** Let L be the finite linear combinations of elements of S, including the zero combination. It is closed under addition and scalar multiplication by combining or scaling coefficients, so it is a subspace containing S. Thus $\mathrm{Span}(S) \subset L$. Every subspace containing S contains all its finite linear combinations, so $L \subset \mathrm{Span}(S)$. Therefore equality holds.

Exercise 30.18: Finite direct sums of vector spaces

Define $V_1 \oplus \cdots \oplus V_n$ as the set of tuples $(v_1, \ldots, v_n)$ with $v_i \in V_i$. Add componentwise and set $c(v_1, \ldots, v_n) = (cv_1, \ldots, cv_n)$. The zero is $(0, \ldots, 0)$ and the inverse is $(-v_1, \ldots, -v_n)$. All vector-space identities hold coordinate by coordinate. For example $c(u+v)$ has coordinate $c(u_i+v_i) = cu_i+cv_i$, the corresponding coordinate of $cu + cv$. The two scalar associativity and distributive laws and the identity law are inherited in the same way. Hence this direct sum is a vector space over F.

Exercise 30.19: The coordinate vector space over an arbitrary field

For F^n use componentwise addition and scalar multiplication. Each vector-space axiom follows from the field laws in each coordinate. The standard vectors e_i with a 1 in coordinate i and zeros elsewhere are a basis:

$$(a_1, \ldots, a_n) = a_1e_1 + \cdots + a_ne_n.$$

This proves spanning, and the coordinates of a zero combination show each $a_i = 0$, proving independence. Thus the dimension is n.

Exercise 30.20: Isomorphism of vector spaces

An isomorphism $T : V \to V'$ over the same field F is a bijection satisfying $T(u + v) = T(u) + T(v)$ and $T(cu) = cT(u)$ for all $u, v \in V, c \in F$. Preserving only the additive group is not enough: the scalar multiplication must also be preserved.

Exercise 30.21: Bases and unique coordinates

Suppose first that $B = \{\beta_1, \ldots, \beta_n\}$ is a basis. Spanning gives an expression for every vector. If $\sum a_i\beta_i = \sum b_i\beta_i$, subtraction yields $\sum(a_i - b_i)\beta_i = 0$. Independence forces $a_i = b_i$ for all i, giving uniqueness. Conversely, if every vector has a unique expression, existence gives spanning. A relation $\sum a_i\beta_i = 0$ is another expression for zero besides the all-zero one, so uniqueness forces every $a_i = 0$. Thus the vectors are independent and form a basis.

Exercise 30.22: Linear systems as a vector-space problem

(a) Let $\alpha_j = (a_{1j}, \ldots, a_{mj})$ be column j. Coordinate i of $\sum_j x_j\alpha_j$ is $\sum_j a_{ij}x_j$. Thus all m scalar equations hold exactly when $\sum_j x_j\alpha_j = \beta = (b_1, \ldots, b_m)$. A solution exists precisely when β belongs to the span of the columns. **(b)** If $m = n$ and the columns form a basis of F^n, every β has a unique basis expansion by Exercise 21. Its coefficients x_j are therefore the unique solution of the system. This argument uses field operations only and is valid in finite characteristic as well.

Exercise 30.23: Coordinates give an isomorphism

Choose a basis $\beta_1, \ldots, \beta_n$ for V. Define $T(\sum a_i\beta_i) = (a_1, \ldots, a_n)$. Uniqueness of basis coordinates makes this well defined and injective; every tuple has the preimage $\sum a_i\beta_i$, so it is onto. Adding vectors or multiplying by a scalar adds or scales their coordinates, respectively. Hence T is an isomorphism of F-vector spaces $V \cong F^n$.

Exercise 30.24: Prescribing a linear transformation on a basis

(a) Every vector has a unique finite expansion $v = \sum_i a_i \beta_i$. Linearity forces $T(v) = \sum_i a_i T(\beta_i)$, so knowing the images of all basis vectors determines every output. **(b)** Conversely prescribe arbitrary $\beta_i' \in V'$ and define $T(\sum_i a_i \beta_i) = \sum_i a_i \beta_i'$. Unique finite expansions make the formula well defined, including for an infinite basis. Combining coefficients proves additivity and homogeneity. The identity from (a) proves no other linear map can have the prescribed basis values. The target vectors may be dependent or repeat.

Exercise 30.25: Kernel and isomorphism for a linear map

(a) A linear transformation is the vector-space analogue of a group or ring homomorphism: it preserves all designated operations. **(b)** Define $\operatorname{Ker} T = \{v : T(v) = 0\}$. It contains zero. For $u, v \in \operatorname{Ker} T$ and $c \in F$, $T(u + v) = 0$ and $T(cu) = 0$. Hence it is a subspace. **(c)** A linear map is an isomorphism onto V' exactly when it is onto and has zero kernel. Indeed $T(u) = T(v)$ is equivalent to $u - v \in \operatorname{Ker} T$, so zero kernel is exactly injectivity. Its inverse then preserves both operations by applying T to their desired identities.

Exercise 30.26: Quotient vector spaces

Use the additive cosets $V/S = \{v + S : v \in V\}$. Define

$$(u + S) + (v + S) = (u + v) + S, \qquad c(v + S) = cv + S.$$

If $u - u', v - v' \in S$, then $(u + v) - (u' + v') \in S$; also $cv - cv' = c(v - v') \in S$. Thus neither operation depends on representatives. The additive group axioms and scalar laws now descend from those of V; for instance $c((u + v) + S) = (cu + cv) + S$. The zero is S and the negative of $v + S$ is $-v + S$. Thus this is a vector space over F.

Exercise 30.27: Image and rank-nullity

(a) The image contains $T(0) = 0$. Its vectors satisfy $T(u)+T(v) = T(u+v)$ and $cT(u) = T(cu)$, so it is a subspace of V'. **(b)** Choose a basis $u_1, \ldots, u_k$ of $\operatorname{Ker} T$, and extend it to a basis $u_1, \ldots, u_k, v_1, \ldots, v_r$ of V. Applying T to basis expansions shows that $T(v_1), \ldots, T(v_r)$ span the image, since each $T(u_i) = 0$. If $\sum_j c_j T(v_j) = 0$, then $\sum_j c_j v_j$ lies in the kernel and can be expressed using the u_i. Independence of the full basis forces every $c_j = 0$. Thus the displayed image vectors form a basis and $\dim \operatorname{im} T = r$. Since $\dim V = k + r$, we obtain $\dim \operatorname{im} T = \dim V - \dim \operatorname{Ker} T$.

31 Algebraic Extensions

Exercise 31.1: A basic quadratic extension

The polynomial $x^2 - 2$ is irreducible over $\mathbb{Q}$, so the degree is two and a basis is $\{1, \sqrt{2}\}$. Spanning follows by reducing powers with $(\sqrt{2})^2 = 2$; independence is equivalent to irrationality of $\sqrt{2}$.

Exercise 31.2: A biquadratic extension

First $[\mathbb{Q}(\sqrt{2}) : \mathbb{Q}] = 2$. The element $\sqrt{3}$ is not in $\mathbb{Q}(\sqrt{2})$: squaring $a + b\sqrt{2} = \sqrt{3}$ gives $a^2 + 2b^2 = 3$, $2ab = 0$. If $b = 0$ this would make 3 a rational square; if $a = 0$ it would make $3/2$ a rational square. Both are impossible by prime factorization of a reduced rational square. Thus adjoining $\sqrt{3}$ has degree two. The tower law gives degree four and the product basis $\{1, \sqrt{2}, \sqrt{3}, \sqrt{6}\}$.

Exercise 31.3: A redundant radical

Since $\sqrt{18} = 3\sqrt{2}$, the extra displayed generator already belongs to $\mathbb{Q}(\sqrt{2}, \sqrt{3})$. Therefore the field is unchanged from Exercise 2: its degree is four and a basis is $\{1, \sqrt{2}, \sqrt{3}, \sqrt{6}\}$.

Exercise 31.4: A cubic radical and a square root

Put $t = \sqrt[3]{2}$. Eisenstein at 2 gives $[\mathbb{Q}(t) : \mathbb{Q}] = 3$. If $\sqrt{3}$ belonged to that field, the tower law would require its degree two to divide three, impossible. Thus $x^2 - 3$ remains irreducible over $\mathbb{Q}(t)$, and the total degree is six. Multiplying the bases gives

$$\{1, t, t^2, \sqrt{3}, t\sqrt{3}, t^2\sqrt{3}\}.$$

Exercise 31.5: Square and cube roots of the same integer

Set $u = \sqrt[6]{2}$. Then $u^3 = \sqrt{2}$ and $u^2 = \sqrt[3]{2}$, while $u = \sqrt{2}/\sqrt[3]{2}$. Hence the given field equals $\mathbb{Q}(u)$. The polynomial $x^6 - 2$ is Eisenstein at 2, so its degree is six and a basis is $\{1, u, u^2, u^3, u^4, u^5\}$.

Exercise 31.6: A sum generating a biquadratic field

Let $s = \sqrt{2} + \sqrt{3}$. Since $(\sqrt{3} - \sqrt{2})s = 1$, the field $\mathbb{Q}(s)$ contains $1/s = \sqrt{3} - \sqrt{2}$. Adding and subtracting gives

$$\sqrt{3} = \frac{s + 1/s}{2}, \qquad \sqrt{2} = \frac{s - 1/s}{2}.$$

Therefore $\mathbb{Q}(s) = \mathbb{Q}(\sqrt{2}, \sqrt{3})$, of degree four. A basis can be $\{1, \sqrt{2}, \sqrt{3}, \sqrt{6}\}$, or equivalently $\{1, s, s^2, s^3\}$ since s has degree four.

Exercise 31.7: A product of square roots

The single generator is $\sqrt{2}\sqrt{3} = \sqrt{6}$, so the field is $\mathbb{Q}(\sqrt{6})$. The polynomial $x^2 - 6$ has no rational root and is irreducible. Thus the degree is two and a basis is $\{1, \sqrt{6}\}$. A product of radicals should not be mistaken for adjoining each radical separately.

Exercise 31.8: Two relatively prime simple degrees

The degrees of $\sqrt{2}$ and $t = \sqrt[3]{5}$ over $\mathbb{Q}$ are two and three. Let $K = \mathbb{Q}(\sqrt{2}, t)$. The tower law says both 2 and 3 divide $[K : \mathbb{Q}]$. On the other hand adjoining t to $\mathbb{Q}(\sqrt{2})$ has degree at most three, so $[K : \mathbb{Q}] \leq 6$. Hence its degree must be six. A basis is $\{1, t, t^2, \sqrt{2}, t\sqrt{2}, t^2\sqrt{2}\}$.

Exercise 31.9: Three displayed cube roots

Put $t = \sqrt[3]{2}$, $u = \sqrt[3]{3}$. Then $\sqrt[3]{6} = tu$ and $\sqrt[3]{24} = 2u$, so the field in the question is exactly $\mathbb{Q}(t, u)$. The proof in Exercise 24 shows $x^3 - 3$ is irreducible over $\mathbb{Q}(t)$. Thus adjoining u has degree three and the tower law gives $[\mathbb{Q}(t, u) : \mathbb{Q}] = 9$. A basis is

$$\{t^i u^j : 0 \leq i \leq 2,\ 0 \leq j \leq 2\}.$$

This is a product of the bases at the two stages. Merely observing that each radical separately has degree three would not by itself establish degree nine.

Exercise 31.10: A quadratic extension over a quadratic base

The top field is $\mathbb{Q}(\sqrt{2}, \sqrt{6}) = \mathbb{Q}(\sqrt{2}, \sqrt{3})$ because $\sqrt{6}/\sqrt{2} = \sqrt{3}$. It has degree four over $\mathbb{Q}$, while the base $\mathbb{Q}(\sqrt{3})$ has degree two. The tower law therefore gives relative degree two. A basis over $\mathbb{Q}(\sqrt{3})$ is $\{1, \sqrt{2}\}$.

Exercise 31.11: A sum over a base containing one summand

Over $F = \mathbb{Q}(\sqrt{3})$, adjoining $\sqrt{2} + \sqrt{3}$ is the same as adjoining $\sqrt{2}$, because we can subtract $\sqrt{3} \in F$. The top field has rational degree four and F degree two. Thus the relative degree is two and a basis is $\{1, \sqrt{2}\}$.

Exercise 31.12: A generator already defining the whole field

Exercise 6 established $\mathbb{Q}(\sqrt{2} + \sqrt{3}) = \mathbb{Q}(\sqrt{2}, \sqrt{3})$. The top and bottom fields here are therefore equal. The relative degree is one, and a basis is $\{1\}$.

Exercise 31.13: A less immediate containment of fields

Write $s = \sqrt{3} + \sqrt{5}$ and $F = \mathbb{Q}(s)$. Since $(\sqrt{5} - \sqrt{3})s = 2$, we recover $\sqrt{5} = (s + 2/s)/2$ and $\sqrt{3} = (s - 2/s)/2$. Thus $F = \mathbb{Q}(\sqrt{3}, \sqrt{5})$, a biquadratic field of rational degree four. The top field is $E = \mathbb{Q}(\sqrt{2}, \sqrt{2}s)$. It contains $s = (\sqrt{2}s)/\sqrt{2}$, so $E = F(\sqrt{2})$. To see that $\sqrt{2} \notin F$, use the independent sign changes of $\sqrt{3}, \sqrt{5}$. If an element of F has square 2, each sign change sends it to itself or its negative. A simultaneous eigenvector for these two sign changes is a rational multiple of exactly one of $1, \sqrt{3}, \sqrt{5}, \sqrt{15}$, as follows by comparing coefficients in their basis. None can square to 2, since $2, 2/3, 2/5, 2/15$ are not rational squares. Thus the relative degree is two and $\{1, \sqrt{2}\}$ is a basis over F.

Exercise 31.14: Correcting algebraic extension

An extension E/F is algebraic when *every element* of E is a zero of a nonzero polynomial in $F[x]$. It need not be generated by finitely many algebraic elements; the field of all algebraic numbers over $\mathbb{Q}$ is a standard infinite-degree example. A finite list of algebraic generators would define a finite extension, which is a stronger condition.

Exercise 31.15: Correcting finite extension

A finite extension E/F means that E is finite dimensional as an F-vector space: $[E : F] < \infty$. Having finitely many field generators is not enough; for example $F(t)$ has one transcendental generator but contains the linearly independent set $1, t, t^2, \ldots$ and has infinite dimension over F.

Exercise 31.16: Algebraic closure inside an extension

The definition is correct: the algebraic closure of F in E is $\overline{F}_E = \{a \in E : a \text{ is algebraic over } F\}$. It is a field because finite sums, products, and inverses of algebraic elements are algebraic. This relative closure need not itself be algebraically closed in the absolute sense.

Exercise 31.17: Correcting algebraically closed

A field F is algebraically closed if every *nonconstant polynomial in* $F[x]$ has a zero in F. Constant nonzero polynomials have no zero in any field, so they must be excluded. By repeatedly factoring off roots, this is equivalent to every nonconstant polynomial splitting into linear factors over F.

Exercise 31.18: A relative closure need not be algebraically closed

Take $F = \mathbb{Q}$ and $E = \mathbb{Q}(\sqrt{2})$, a proper algebraic extension. Every element of E is algebraic over F, so $\overline{F}_E = E$. But E is not algebraically closed: it is a real field and $x^2 + 1$ has no zero in it. This separates the relative and absolute uses of closure.

Exercise 31.19: True and false statements

(a) False. $\mathbb{Q}(\sqrt{2})/\mathbb{Q}$ has degree two but infinitely many elements.

(b) True. In an n-dimensional extension, $1, a, \ldots, a^n$ are dependent for every a.

(c) False. The field of all algebraic numbers over $\mathbb{Q}$ has unbounded element degrees; Exercise 36 gives a proof.

(d) True. Repeated application of the tower law multiplies finitely many finite degrees.

(e) False. $\sqrt{2} \in \mathbb{R}$ is algebraic over $\mathbb{Q}$ and is not rational.

(f) True. Any element algebraic over the already algebraically closed $\mathbb{C}$ must be a root of a linear factor, hence belongs to $\mathbb{C}$.

(g) False. x has no square root in $\mathbb{C}(x)$: a putative rational-function square would give an even exponent of the factor x equal to an odd exponent.

(h) False. Every field has an algebraic closure, including the rational-function field $\mathbb{C}(x)$; containing $\mathbb{C}$ does not supply roots involving the new indeterminate.

(i) False. An algebraic closure of $\mathbb{Z}_p$ is algebraically closed and still has characteristic p.

(j) False. $\mathbb{C}$ is algebraically closed and extends $\mathbb{Q}$, but contains the transcendental number π.

Exercise 31.20: Synopsis of finite implies algebraic

If $[E : F] = n$, the $n + 1$ vectors $1, a, \ldots, a^n$ are linearly dependent over F, providing a nonzero polynomial in $F[x]$ that vanishes at the arbitrary element $a \in E$.

Exercise 31.21: Synopsis of the tower law

Multiply each vector in an F-basis of E by each vector in an E-basis of K. Expanding first over E and then over F proves that these products span K, and grouping a proposed relation by the E-basis and using independence at both stages proves they are independent. Thus their number is $[K : E][E : F]$.

Exercise 31.22: Every nonreal complex number generates over the reals

For $z = a + bi$ with real a, b and $b \neq 0$, the field $\mathbb{R}(z)$ contains $(z - a)/b = i$. It therefore contains $\mathbb{R}(i) = \mathbb{C}$. The reverse inclusion holds because $z \in \mathbb{C}$, so $\mathbb{R}(z) = \mathbb{C}$. Division by b is legitimate because it is a nonzero element of the base field.

Exercise 31.23: Prime-degree extensions are simple

Take $a \in E \setminus F$. The intermediate field $F(a)$ satisfies

$$[E : F] = [E : F(a)][F(a) : F].$$

The left side is prime and the second factor is greater than one because $a \notin F$. Hence $[F(a) : F] = [E : F]$ and $[E : F(a)] = 1$. Thus $E = F(a)$ for every such a, proving simplicity.

Exercise 31.24: A cubic stays irreducible over a pure cubic field

Put $t = \sqrt[3]{2}$ and $K = \mathbb{Q}(t)$. A reducible cubic $x^3 - 3$ over K would have a zero in K. Since $K \subset \mathbb{R}$, that zero must be $u = \sqrt[3]{3}$. Suppose $u = a + bt + ct^2$ with $a, b, c \in \mathbb{Q}$. The rational polynomial $H(X) = (a + bX + cX^2)^3 - 3$ vanishes at t, so its minimal polynomial $X^3 - 2$ divides $H(X)$. Let ω be a nonreal cube root of unity. All three roots $t, \omega t, \omega^2 t$ of $X^3 - 2$ therefore make H vanish. The three values $a + bt + ct^2$, $a + b\omega t + c\omega^2 t^2$, and $a + b\omega^2 t + c\omega t^2$ are consequently cube roots of three. They are distinct. Indeed u has degree three over $\mathbb{Q}$ by Eisenstein at three. Under the assumption $u \in K$, the tower law gives $\mathbb{Q}(u) = K$. Substituting any of the three roots for t defines a field embedding of K into $\mathbb{C}$, because $K \cong \mathbb{Q}[X]/(X^3 - 2)$. If two such embeddings had the same value at u, they would

agree on all of $\mathbb{Q}(u) = K$, including t, contradicting their distinct assigned values of t. Hence the three values are exactly $u, \omega u, \omega^2 u$. Their sum is zero, but summing the three substituted expressions gives $3a$, so $a = 0$. The sum of their squares is also zero; with $a = 0$, summing $(bt + ct^2)^2$ over the three substitutions leaves $6bc\,t^3 = 12bc$. Thus $bc = 0$. If $c = 0$, cubing gives $b^3 = 3/2$; if $b = 0$, cubing gives $c^3 = 3/4$. Neither rational number is a rational cube, by its prime exponents. Both cases are impossible. Thus $u \notin K$, and the cubic is irreducible.

Exercise 31.25: Degrees arising from repeated square roots

Adjoining a square root of a nonsquare has degree exactly two. A tower of r such steps therefore has total degree 2^r by the tower law. Any element in that tower has degree over the original field dividing 2^r, hence itself a power of two. The polynomial $x^{14} - 3x^2 + 12$ is Eisenstein at 3: every nonleading coefficient is divisible by 3 and the constant 12 is not divisible by 9. It is irreducible over $\mathbb{Q}$, so each of its zeros has degree fourteen. Since fourteen is not a power of two, no zero can belong to a finite tower of quadratic extensions. In particular no expression made from finitely many rational functions and nested square roots of rationals can give such a zero.

Exercise 31.26: An intermediate domain in a finite extension

Let $0 \neq d \in D$. It is algebraic over F because $D \subset E$ and E/F is finite. Its minimal polynomial has nonzero constant term: otherwise it would have a factor x, contradicting irreducibility for a nonzero root. Write

$$d^n + a_{n-1}d^{n-1} + \cdots + a_1 d + a_0 = 0, \qquad a_0 \neq 0.$$

Rearrange to obtain

$$d^{-1} = -a_0^{-1}(d^{n-1} + a_{n-1}d^{n-2} + \cdots + a_1) \in D.$$

The coefficients lie in $F \subset D$ and D is closed under its ring operations. Thus every nonzero element of D has its inverse in D, proving it is a field.

Exercise 31.27: Recovering two quadratic radicals from their sum

Let $s = \sqrt{3} + \sqrt{7}$. The inclusion $\mathbb{Q}(s) \subset \mathbb{Q}(\sqrt{3}, \sqrt{7})$ is immediate. Since $s \neq 0$ and $(\sqrt{7} - \sqrt{3})s = 4$, the first field contains $4/s = \sqrt{7} - \sqrt{3}$. Consequently

$$\sqrt{7} = \frac{s + 4/s}{2}, \qquad \sqrt{3} = \frac{s - 4/s}{2}.$$

It contains both generators of the second field, yielding the reverse inclusion and equality.

Exercise 31.28: The general sum-of-two-square-roots identity

Put $s = \sqrt{a} + \sqrt{b} \neq 0$. Then

$$\frac{a - b}{s} = \sqrt{a} - \sqrt{b}.$$

Thus

$$\sqrt{a} = \frac{1}{2}\left(s + \frac{a - b}{s}\right), \qquad \sqrt{b} = \frac{1}{2}\left(s - \frac{a - b}{s}\right).$$

Since $a, b \in \mathbb{Q}$ and s is nonzero, both expressions belong to $\mathbb{Q}(s)$. This proves $\mathbb{Q}(\sqrt{a}, \sqrt{b}) \subset \mathbb{Q}(s)$; the reverse inclusion is immediate. The formula remains valid when $a = b$ provided the stated nonzero-sum condition holds.

Exercise 31.29: A divisibility obstruction to roots

Suppose $a \in E$ were a root of the irreducible polynomial $p \in F[x]$. Its minimal polynomial is the monic associate of p, so $[F(a) : F] = \deg p$. The tower law would then give $[E : F] = [E : F(a)] \deg p$, making $\deg p$ a divisor of $[E : F]$. This contradicts the hypothesis. Therefore no zero lies in E.

Exercise 31.30: Squaring an odd-degree element

The element α^2 lies in the finite extension $F(\alpha)$, so is algebraic over F. Moreover α satisfies $x^2 - \alpha^2$ over $F(\alpha^2)$, so $[F(\alpha) : F(\alpha^2)]$ is one or two. The tower law makes this degree divide the odd integer $[F(\alpha) : F]$, forcing it to be one. Hence $F(\alpha) = F(\alpha^2)$, and α^2 has the same odd degree as α.

Exercise 31.31: Transitivity of arbitrary algebraic extensions

If K/F is algebraic, every element of $E \subset K$ is algebraic over F, and every annihilating polynomial over F is also one over E. Thus E/F and K/E are algebraic. Conversely suppose both intermediate extensions are algebraic, and take $a \in K$. A polynomial annihilating a over E has only finitely many coefficients $c_0, \ldots, c_n$. They are algebraic over F, so $L = F(c_0, \ldots, c_n)$ is a finite extension of F by repeatedly adjoining one algebraic element. The element a is algebraic over L, hence $L(a)/L$ is finite. The finite tower law shows $L(a)/F$ is finite, so a is algebraic over F. This proves transitivity without assuming the original extensions finite.

Exercise 31.32: Elements outside the relative algebraic closure

Let $A = \overline{F}_E$. This field is algebraic over F by its definition. If $a \in E$ were algebraic over A, transitivity from Exercise 31 would make a algebraic over F, and hence place it in A. Therefore every $a \in E \setminus A$ must be transcendental over A.

Exercise 31.33: Relative closure inside an algebraically closed field

Let $A = \overline{F}_E$, where E is algebraically closed. Take any nonconstant $p \in A[x]$. Since $p \in E[x]$, it has a zero $a \in E$. This a is algebraic over A, and A/F is algebraic, so transitivity makes a algebraic over F. Hence $a \in A$ by the definition of A. Thus every nonconstant polynomial over A has a zero there, so A is algebraically closed.

Exercise 31.34: Containing all roots of base-field polynomials

Work in a common algebraic closure containing the given algebraic extensions. Every element of an algebraic closure $\overline{F}$ is a zero of a nonzero polynomial in $F[x]$, so the hypothesis implies $\overline{F} \subset E$. Any $a \in E$ is algebraic over F, hence also over $\overline{F}$. Its polynomial over the algebraically closed field $\overline{F}$ factors into linear factors, so a must belong to $\overline{F}$. Therefore $E = \overline{F}$ and is algebraically closed.

Equivalently, the hypothesis that all algebraic roots have already been included leaves no possible further algebraic element.

Exercise 31.35: A finite field is not algebraically closed

Suppose $|F| = q$ and its characteristic is odd. Squaring pairs the nonzero elements as $\{a, -a\}$, with distinct members, and equal squares arise precisely from those pairs. There are $(q-1)/2$ nonzero squares, so choose a nonsquare $d \in F$. Then $x^2 - d$ has no zero in F, proving F is not algebraically closed. For completeness, in any finite field every a satisfies $a^q = a$, so $x^q - x - 1$ evaluates to $-1 \neq 0$ at every element. This proves the parenthetical characteristic-two claim as well.

Exercise 31.36: The algebraic numbers have infinite degree

For every positive integer n, the polynomial $x^n - 2$ is Eisenstein at 2, so a chosen root has degree n over $\mathbb{Q}$. All these roots lie in the algebraic closure of $\mathbb{Q}$ in $\mathbb{C}$. If that closure had finite degree d, every element degree would divide d and in particular be at most d. Taking $n > d$ contradicts this. Thus the algebraic closure is not a finite extension of $\mathbb{Q}$.

Exercise 31.37: Finite extensions of the reals

A real polynomial factors over $\mathbb{R}$ into linear and irreducible quadratic factors, by pairing conjugate roots in its complex factorization. Thus every irreducible polynomial over $\mathbb{R}$ has degree one or two. If a finite extension $E/\mathbb{R}$ has no element outside $\mathbb{R}$, it is $\mathbb{R}$. Otherwise choose $a \in E \setminus \mathbb{R}$. Its minimal polynomial is an irreducible quadratic $x^2 + bx + c$ with $c - b^2/4 > 0$. The element $j = (a + b/2)/\sqrt{c - b^2/4}$ satisfies $j^2 = -1$, so $\mathbb{R}(a) = \mathbb{R}(j) \cong \mathbb{C}$. This intermediate field is algebraically closed, while E is algebraic over it. An algebraically closed field has no proper algebraic extension: the minimal polynomial of any algebraic element must be linear. Hence $E = \mathbb{R}(a) \cong \mathbb{C}$.

Exercise 31.38: Maximal ideals from Zorn lemma

Let I be the given proper ideal, and partially order the proper ideals containing I by inclusion. The family is nonempty because it contains I. For any nonempty chain, let J be its union. Two elements of J lie together in one chain member, so their difference lies in J; absorption follows within the member containing the chosen element. Thus J is an ideal containing I. It is proper: if $1 \in J$, then 1 lies in one chain member, which would be the whole ring, contradicting its propriety. The empty chain has upper bound I. Zorn's lemma now yields a maximal member M. Any proper ideal properly containing M would still contain I, contrary to maximality in this ordered family. Hence M is a maximal ideal of R containing I.

32 Geometric Constructions

Exercise 32.1: Deriving the cosine triple-angle identity

Euler's formula gives $e^{3i\theta} = (\cos\theta + i\sin\theta)^3$. Its real part is

$$\cos 3\theta = \cos^3\theta - 3\cos\theta\sin^2\theta.$$

Substitute $\sin^2\theta = 1 - \cos^2\theta$ and collect terms:

$$\cos 3\theta = \cos^3\theta - 3\cos\theta + 3\cos^3\theta = 4\cos^3\theta - 3\cos\theta.$$

The imaginary terms in the cube do not contribute to the real part.

Exercise 32.2: True and false statements

(a) True for a nondegenerate cube. If its constructible edge is $a > 0$ and the doubled cube had constructible edge b, then $b/a = \sqrt[3]{2}$ would be constructible, contrary to its degree three.

(b) True. Since even a unit cube cannot be doubled, no procedure can double every constructible cube; in fact (a) gives the stronger individual impossibility.

(c) True for a nonzero radius. Squaring a circle of constructible radius r would construct a square edge $r\sqrt{\pi}$; division by r and squaring would make π constructible and therefore algebraic, a contradiction.

(d) False. A right angle can be trisected into 30° angles, which are constructible.

(e) True. A constructible number lies in a tower of quadratic extensions, so its degree divides a power of two and is itself a power of two.

(f) False. Only a necessary degree condition was proved; being of power-of-two degree alone does not establish membership in a tower of quadratic extensions.

(g) True. The final obstruction uses that a degree divisible by three cannot divide a power of two, a consequence of prime factorization.

(h) True as the intended methodological observation: counting dimensions rules out constructions without trying all diagrams.

(i) True. Each individual constructible number has a finite construction by definition.

(j) False. A finite sequence gives only finitely many constructed points or segments, while the constructible numbers already include infinitely many integers. There is no common finite construction listing all of them.

Exercise 32.3: Why a regular nonagon is not constructible

If a regular 9-gon were constructible, its central angle 40° would be constructible. Let $u = 2\cos 40^\circ$. Since $3 \cdot 40^\circ = 120^\circ$, the triple-angle identity yields

$$u^3 - 3u = 2\cos 120^\circ = -1, \qquad u^3 - 3u + 1 = 0.$$

The monic cubic has only possible rational roots ± 1, and neither is a root. Thus it is irreducible over $\mathbb{Q}$ and u has degree three. Every constructible number has degree a power of two, so u and the regular 9-gon are not constructible.

Exercise 32.4: Constructing thirty degrees algebraically

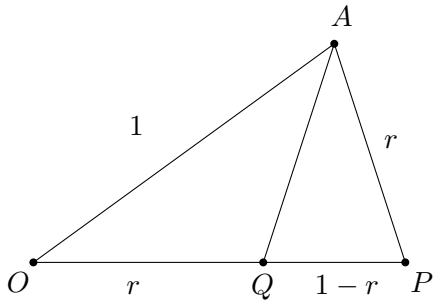
The numbers $\cos 30^\circ = \sqrt{3}/2$ and $\sin 30^\circ = 1/2$ are constructible: constructible numbers include rationals and are closed under positive square roots. Hence the unit-circle point $(\sqrt{3}/2, 1/2)$ is constructible, and the ray from the origin through it makes a 30° angle with the positive horizontal axis. This gives the requested algebraic justification.

Exercise 32.5: The golden-ratio chord of a decagon

In the source triangle, $OA = OP = 1$, $\angle O = 36^\circ$, and $AP = r$. Thus the base angles at A, P are 72°. The bisector AQ gives $\angle QAP = 36^\circ$, so $\triangle OAP \sim \triangle APQ$ with correspondence $O \leftrightarrow A$, $A \leftrightarrow P$, $P \leftrightarrow Q$. Also $AQ = AP = r$, and $OQ = AQ = r$, so $PQ = 1 - r$. The similar triangles give $1/r = r/(1-r)$, hence $r^2 + r - 1 = 0$. Positivity yields

$$r = \frac{\sqrt{5} - 1}{2},$$

a constructible length. A chord of this length in a unit circle subtends 36°; the law of cosines equivalently gives $\cos 36^\circ = 1 - r^2/2 = (1 + \sqrt{5})/4$. Thus one can construct the decagon by stepping this chord around the circle. Joining alternate vertices gives a regular pentagon.



Exercise 32.6: Constructing a regular twenty-gon

Exercise 5 constructs the central angle 36° of a decagon. Bisect it to obtain $18^\circ = 360^\circ/20$, using the standard compass-and-straightedge angle-bisector construction. Repeatedly mark this angle around a circle to obtain the twenty equally spaced vertices. Thus the regular 20-gon is constructible.

Exercise 32.7: Constructing a regular thirty-gon

A regular hexagon gives 60° and a decagon gives 36°. Subtracting constructible angles gives 24°, then bisecting gives $12^\circ = 360^\circ/30$. Both subtraction and bisection are standard straightedge-and-compass constructions. Marking this central angle around a circle constructs the regular 30-gon.

Exercise 32.8: Trisecting seventy-two degrees

The angles 60° and 36° are constructible, so their difference 24° is constructible. Since $72^\circ = 3 \cdot 24^\circ$, this gives a trisector. For the given 72° angle, copy 24° from one side twice inside it. Thus some angles that are readily constructible can also be trisected, despite the impossibility of a universal trisection procedure.

Exercise 32.9: Constructing a regular fifteen-gon

Subtract the decagon angle 36° from the hexagon angle 60° to construct 24°. This is $360^\circ/15$, so fifteen equal successive central angles locate the vertices of a regular 15-gon. Equivalently alternate vertices of the constructible regular 30-gon give it.

Exercise 32.10: A brief explanation for a geometry teacher

Every coordinate obtainable by compass and straightedge is built from rational numbers using arithmetic and repeated square roots. Consequently its algebraic degree must be a power of two. If 60° could be trisected, the constructible number $u = 2\cos 20^\circ$ would satisfy $u^3 - 3u - 1 = 0$, an irreducible cubic since neither possible rational root 1 nor -1 works. Its degree would be three, contradicting the power-of-two condition.

33 Finite Fields

Exercise 33.1: A power of two

The factorization is $4096 = 2^{12}$. For every prime p and positive integer n, there exists a finite field of order p^n. Thus a field with 4096 elements exists, namely $\mathrm{GF}(2^{12})$.

Exercise 33.2: An integer with two distinct prime factors

We have $3127 = 53 \cdot 59$. Both factors are prime and distinct, so this is not a prime power. Every finite field has prime-power order; therefore no field of 3127 elements exists.

Exercise 33.3: A cube of a prime

Compute $41^2 = 1681$ and $1681 \cdot 41 = 68{,}921$. Since 41 is prime, $68{,}921 = 41^3$ is a prime power, so $\mathrm{GF}(41^3)$ exists and has the required number of elements.

Exercise 33.4: Primitive eighth roots

The nonzero elements of GF(9) form a cyclic group of order 8. A primitive eighth root is exactly a generator of this group. A cyclic group of order eight has $\varphi(8) = 4$ generators, corresponding to powers with exponents $1, 3, 5, 7$ relatively prime to eight. Thus the answer is four.

Exercise 33.5: Primitive eighteenth roots

The multiplicative group of GF(19) is cyclic of order 18. Its elements of order 18 are its $\varphi(18) = 18(1 - 1/2)(1 - 1/3) = 6$ generators. Hence there are six primitive eighteenth roots of unity.

Exercise 33.6: Primitive fifteenth roots

The multiplicative group of GF(31) is cyclic of order 30. It has a unique subgroup of order 15, whose generators are exactly the elements of order 15 in the whole group. Their number is $\varphi(15) = 15(1 - 1/3)(1 - 1/5) = 8$. Thus the answer is eight.

Exercise 33.7: An order absent from the multiplicative group

The nonzero elements of GF(23) form a group of order 22. An element's order must divide 22, but 10 does not. Therefore there are no elements of order 10, and hence zero primitive tenth roots of unity. Merely computing $\varphi(10)$ without checking divisibility would give an incorrect answer.

Exercise 33.8: True and false statements

(a) True. The multiplicative group of a finite field is cyclic.

(b) False. The additive group of $\mathrm{GF}(p^n)$ is $(\mathbb{Z}_p)^n$; for $n > 1$ every nonzero element has order p, so it is not cyclic.

(c) True. The complex zeros of $x^{28} - 1$ are $e^{2\pi ik/28}$ and form the cyclic group generated by $e^{2\pi i/28}$.

(d) False. $60 = 2^2 \cdot 3 \cdot 5$ is not a prime power.

(e) True. $125 = 5^3$ is a prime power.

(f) False. $36 = 2^2 \cdot 3^2$ is not a prime power.

(g) True. The powers $i, i^2 = -1, i^3 = -i, i^4 = 1$ show that i has order four.

(h) True. Over every finite field there is an irreducible polynomial of each positive degree, including degree 58 over $\mathbb{Z}_2$.

(i) False. Any proposed rational generator has only finitely many primes in its numerator and denominator; its integer powers cannot produce a prime absent from that list. Thus $\mathbb{Q}^*$ is not cyclic.

(j) True. An isomorphic image has the same finite size as F, and within the fixed algebraic closure there is exactly one subfield of that size, the roots of $x^{|F|} - x$. The image is therefore F itself.

Exercise 33.9: Two cubic generators inside one algebraic closure

Both binary cubics $x^3 + x^2 + 1$ and $x^3 + x + 1$ have value 1 at $0, 1$, so both are irreducible. Thus $\mathbb{Z}_2(\alpha)$ and $\mathbb{Z}_2(\beta)$ each have eight elements. Every element of either field satisfies $x^8 - x = 0$. In the common algebraic closure this polynomial has at most eight roots, and each of these fields already consists of eight roots. They must therefore be the same set, proving equality of fields rather than just an abstract isomorphism.

Exercise 33.10: Irreducibles divide a Frobenius polynomial

Let $f \in \mathbb{Z}_p[x]$ be irreducible of degree n, replacing it by a monic associate if necessary. The quotient $E = \mathbb{Z}_p[x]/(f)$ is a field of p^n elements. For $\alpha = x + (f)$, Lagrange's theorem in E^* (and the zero case) gives $\alpha^{p^n} = \alpha$. The minimal polynomial of α is the monic associate of f, so it divides every polynomial vanishing at α, in particular $x^{p^n} - x$. Hence f divides that polynomial up to its harmless unit coefficient.

Exercise 33.11: A multiplicative generator has full degree

Every nonzero element of F is a power of α, so belongs to the subfield $\mathbb{Z}_p(\alpha)$. Zero also belongs, hence $\mathbb{Z}_p(\alpha) = F$. If $d = \deg(\alpha, \mathbb{Z}_p)$, the simple extension has p^d elements. Comparing with $|F| = p^n$ gives $d = n$.

Exercise 33.12: Subfields of a finite field

Let $E = \mathrm{GF}(p^n)$ and $m \mid n$. Set $K = \{a \in E : a^{p^m} = a\}$. Iterated Frobenius gives $(a+b)^{p^m} = a^{p^m} + b^{p^m}$ and $(ab)^{p^m} = a^{p^m} b^{p^m}$, so K is closed under addition, multiplication, and negatives; inverses of nonzero

fixed elements are fixed too. It contains $0, 1$, hence is a subfield. Because $p^m - 1$ divides $p^n - 1$, the cyclic group E^* has exactly $p^m - 1$ solutions of $a^{p^m-1} = 1$. Including zero gives $|K| = p^m$. Any subfield of size p^m consists of roots of $x^{p^m} - x$, so must equal this K by counting. This proves uniqueness. Conversely any subfield has size p^m, and the tower law makes m divide n.

Exercise 33.13: Factoring a Frobenius polynomial

The derivative of $x^{p^n} - x$ is -1, so it has no repeated irreducible factor. An irreducible monic polynomial f of degree d divides it exactly when its roots lie in $\mathrm{GF}(p^n)$. If such a root a lies there, $\mathbb{Z}_p(a)$ is a subfield of order p^d, so $d \mid n$ by the tower law. Conversely if $d \mid n$, the field $\mathbb{Z}_p(a)$ of p^d elements embeds as the unique subfield of $\mathrm{GF}(p^n)$ of that size; all its elements satisfy $x^{p^n} = x$, so f divides the polynomial. Thus its monic irreducible factors are exactly those whose degrees divide n, each appearing once. Since the polynomial is monic, the product has no additional nonunit constant.

Exercise 33.14: Euler criterion and a quadratic over seventeen elements

(a) Choose a generator g of $\mathbb{Z}_p^*$, which has even order $p - 1$. Write the nonzero residue $a = g^k$. It is a square exactly when k is even: the squares are the even powers of g. Also

$$a^{(p-1)/2} = 1 \quad \Longleftrightarrow \quad (p-1) \mid k(p-1)/2 \quad \Longleftrightarrow \quad 2 \mid k.$$

This proves the criterion, including both implications. **(b)** Modulo seventeen, $6^2 \equiv 2$, $6^4 \equiv 4$, and $6^8 \equiv 16 = -1 \neq 1$. Hence six is not a quadratic residue, so $x^2 - 6$ has no root in $\mathbb{Z}_{17}$ and is irreducible there.